

Лабораторная работа

Номер 15

Кобзев Д. К.

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Российский университет дружбы народов, Москва, Россия

Информация

- Кобзев Дмитрий Константинович
- Студент
- Российский университет дружбы народов
- НПИбд-01-23

Целью данной работы является настройка динамической маршрутизации между территориями организации.

Настраиваем маршрутизатор msk-donskaya-gw-1 (Рис. 1.1).

```
msk-donskaya-dkkobzev-gw-1>en
Password:
msk-donskaya-dkkobzev-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-dkkobzev-gw-1(config)#router ospf 1
msk-donskaya-dkkobzev-gw-1(config-router)#router-id 10.128.254.1
msk-donskaya-dkkobzev-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
msk-donskaya-dkkobzev-gw-1(config-router)#exit
```

Рис. 1: Настройка маршрутизатора msk-donskaya-gw-1

Проверяем состояние протокола OSPF на маршрутизаторе msk-donskaya-gw-1 (Рис. 1.2).

```
msk-donskaya-dikobnev-gw-1#sh ip ospf
Routing Process "ospf 1" with ID 10.128.254.1
Supports only single TOS(TO00) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs, Minimum LSA arrival 1 secs
Number of external LSA 0, Checksum sum 0x000000
Number of opaque AS LSA 0, Checksum sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DCbitless external and opaque AS LSA 0
Number of areas in this router is 1, 1 normal 0 stub 0 nssa
External flood list length 0
Area BACKBONE(0)
Number of interfaces in this area is 5
Area has no authentication
SPF algorithm executed 1 times
Area ranges are
Number of LSA 1, Checksum sum 0x003112a
Number of opaque link LSA 0, Checksum sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0

msk-donskaya-dikobnev-gw-1#
msk-donskaya-dikobnev-gw-1#sh ip ospf neighbor

msk-donskaya-dikobnev-gw-1#sh ip route
Codes: L - local, C - connected, N - static, H - N3, M - mobile, W - WSP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
NI - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, I - IGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
p - periodic downloaded static route

Gateway of last resort is 190.51.100.1 to network 0.0.0.0

O 10.0.0.0/8 is variably subnetted, 18 subnets, 6 masks
C 10.120.0.0/24 is directly connected, FastEthernet0/0.3
C 10.120.0.1/32 is directly connected, FastEthernet0/0.3
C 10.120.1.0/24 is directly connected, FastEthernet0/0.2
C 10.120.1.1/32 is directly connected, FastEthernet0/0.2
C 10.120.3.0/24 is directly connected, FastEthernet0/0.101
C 10.120.3.1/32 is directly connected, FastEthernet0/0.101
C 10.120.4.0/24 is directly connected, FastEthernet0/0.102
C 10.120.4.1/32 is directly connected, FastEthernet0/0.102
C 10.120.5.0/24 is directly connected, FastEthernet0/0.103
C 10.120.5.1/32 is directly connected, FastEthernet0/0.103
C 10.120.6.0/24 is directly connected, FastEthernet0/0.104
C 10.120.6.1/32 is directly connected, FastEthernet0/0.104
C 10.120.255.0/30 is directly connected, FastEthernet0/1.6
C 10.120.255.1/32 is directly connected, FastEthernet0/1.6
C 10.120.255.4/30 is directly connected, FastEthernet0/1.6
C 10.120.255.5/32 is directly connected, FastEthernet0/1.6
O 10.129.0.0/16 [1/0] via 10.128.255.2
O 10.130.0.0/16 [1/0] via 10.128.255.6
O 190.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
```

Рис. 2: Проверка состояния протокола OSPF на маршрутизаторе msk-donskaya-gw-1

Настраиваем маршрутизатор msk-q42-gw-1 (Рис. 1.3).

```
msk-q42-dkkobzev-gw-1>en
Password:
msk-q42-dkkobzev-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-dkkobzev-gw-1(config)#router ospf 1
msk-q42-dkkobzev-gw-1(config-router)#router-id 10.128.254.2
msk-q42-dkkobzev-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
msk-q42-dkkobzev-gw-1(config-router)#exit
```

Рис. 3: Настройка маршрутизатора msk-q42-gw-1

Настраиваем маршрутизирующий коммутатор msk-hostel-gw-1 (Рис. 1.4)

```
msk-hostel-dkkobzev-gw-1>en
Password:
msk-hostel-dkkobzev-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-hostel-dkkobzev-gw-1(config)#router ospf 1
msk-hostel-dkkobzev-gw-1(config-router)#router-id 10.128.254.3
msk-hostel-dkkobzev-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
msk-hostel-dkkobzev-gw-1(config-router)#exit
```

Рис. 4: Настройка маршрутизирующего коммутатора msk-hostel-gw-1

Настраиваем маршрутизатор sch-sochi-gw-1 (Рис. 1.5).

```
sch-sochi-dkkobzev-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-dkkobzev-gw-1(config)#router ospf 1
sch-sochi-dkkobzev-gw-1(config-router)#router-id 10.128.254.4
sch-sochi-dkkobzev-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
sch-sochi-dkkobzev-gw-1(config-router)#exit
```

Рис. 5: Настройка маршрутизатора sch-sochi-gw-1

Настраиваем интерфейсы коммутатора provider-sw-1 (Рис. 1.6).

```
provider-dkkobzev-sw-1>en
Password:
provider-dkkobzev-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
provider-dkkobzev-sw-1(config)#vlan 7
provider-dkkobzev-sw-1(config-vlan)#name q42-sochi
provider-dkkobzev-sw-1(config-vlan)#exit
provider-dkkobzev-sw-1(config)#int vlan7
provider-dkkobzev-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan7, changed state to up

provider-dkkobzev-sw-1(config-if)#no shutdown
provider-dkkobzev-sw-1(config-if)#exit
```

Рис. 6: Настройка интерфейсов коммутатора provider-sw-1

Настраиваем маршрутизатор msk-q42-gw-1 (Рис. 1.7).

```
msk-q42-dkkobzev-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-dkkobzev-gw-1(config)#int f0/1.7
msk-q42-dkkobzev-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.7, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.7, changed state to up

msk-q42-dkkobzev-gw-1(config-subif)#encapsulation dot1Q 7
msk-q42-dkkobzev-gw-1(config-subif)#ip address 10.128.255.9 255.255.255.252
msk-q42-dkkobzev-gw-1(config-subif)#description sochi
msk-q42-dkkobzev-gw-1(config-subif)#exit
```

Рис. 7: Настройка маршрутизатора msk-q42-gw-1

Настраиваем коммутатор sch-sochi-sw-1 (Рис. 1.8).

```
sch-sochi-dkkobzev-sw-1>en
Password:
sch-sochi-dkkobzev-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-dkkobzev-sw-1(config)#vlan 7
sch-sochi-dkkobzev-sw-1(config-vlan)#name q42-sochi
sch-sochi-dkkobzev-sw-1(config-vlan)#exit
sch-sochi-dkkobzev-sw-1(config)#int vlan7
sch-sochi-dkkobzev-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan7, changed state to up

sch-sochi-dkkobzev-sw-1(config-if)#no shutdown
sch-sochi-dkkobzev-sw-1(config-if)#exit
```

Рис. 8: Настройка коммутатора sch-sochi-sw-1

Настраиваем маршрутизатор sch-sochi-gw-1 (Рис. 1.9).

```
sch-sochi-dkkobzev-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-dkkobzev-gw-1(config)#int f0/0.7
sch-sochi-dkkobzev-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.7, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.7, changed state to up

sch-sochi-dkkobzev-gw-1(config-subif)#encapsulation dot1Q 7
sch-sochi-dkkobzev-gw-1(config-subif)#ip address 10.128.255.10 255.255.255.252
sch-sochi-dkkobzev-gw-1(config-subif)#description q42
sch-sochi-dkkobzev-gw-1(config-subif)#exit
```

Рис. 9: Настройка маршрутизатора sch-sochi-gw-1

В результате выполнения лабораторной работы мною была настроена динамическая маршрутизация между территориями организации.